

Project N°	Unit	Doc.Type	Material Code	Serial N°	Rev. index
	032	PDS	0910	0025	1

		T811 BOTTOM LUBES PRODUCT PUMPS		Doc. No.: Page 3 of 4	
				Client No.: Rev.: 1	
		P818 (A/B)		Date: 17-Dec-2025	
		Service: T811 BOTTOM LUBES PRODUCT PUMPS			
		Client:		Job No.: -	
		Project:			
		Location:			
		Unit: Hydrocracking Unit		Area: 032	

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2	ITEM NORMAL OPERATION				P818 (A/B) (9)
3	SPARE ITEM				-
4					
5	NUMBER				TWO (1W+1S) (20)
6	PUMP TYPE				Centrifugal Pump
7	SERVICE				Lubes Product
8					
9	PRODUCT HANDLED	Product	Lubes (Hydrocarbon)		
10		Auto-ignition temperature	°F	665.0	
11		Corrosive/Erosive	No/No		
12		Normal pumping temp. Tp	°F	529.58	
13		Viscosity at Tp	cP	0.26	
14		Vapor pressure at Tp	psia	3.22	
15		Specific gravity at Tp	0.671		
16		Density at Tp	lb/ft³	41.91	
17		WT% Solids	-		
18					
19	PUMP	Normal flowrate (11)	GPM	344	
20		Minimum flowrate (12)	GPM	172	
21		Design flowrate	GPM	378	
22					
23		Discharge pressure	psig	155.0	
24		Suction pressure	psig	-2.9	
25		Differential pressure at design flow	psi	158.0	
26		Differential head (A)	ft	542.9	
27		Available NPSH	ft	26.5 (7)	
28		Normal shaft power (est.)	hp	50.9 (3)	
29		Normal electrical consumption (est.)	hp	By Vendor (3)	
30					
31		Design temperature	°F	650 (2) ⚠	
32		Suction design pressure	psig	111.7	
33		Material of construction (Casing / Impeller)	CS / CS		
34	Driver type : normal		Electric Motor		
35	spare		Electric Motor		
36	Speed control	Yes/No	No		
37	speed control by :				
38	Turbine speed control	Manual/Auto	N/A		
39	Expected speed range		-		
40	Open valve start-up	Yes/No	Yes (10) ⚠		
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42	GENERAL NOTE				
43	(A) The choice of a pump having a max. head more than 1.2 time the design head shall be submitted to the Process Dept approval.				
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PROCESS DATA SHEET

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	T811 BOTTOM LUBES PRODUCT PUMPS	Doc. No.:		Page	4 of 4
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		Unit:	Hydrocracking Unit	Area:	032

GENERAL NOTES:

(B) For utility conditions and site data, please refer to Process Design Basis. For utility condition at vendor battery limit, vendor to consider 7.11 psia additional pressure drop to arrive at vendor battery limit pressure (stringent among min/max/nor to be considered).

(C) Vendor to provide utility consumption summary with following minimum data per item: Name of utility, consumption rate/quantity, intermittent/continuous, frequency if intermittent, and required operating min/max pressure and temperature.

(D) Vendor to provide all the auxiliary instrument and equipment datasheet as applicable. Vendor to ensure units of measurement in datasheets as per Process Design Basis.

(E) For mechanical seal/bearing cooling systems, where cooling water is used, maximum pressure drop allowed across supply and return at pump skid battery limit is 9.95 psia. Vendor shall ensure backflush provision and a bleed point with valve and blind at upsetstream of isolation valve on cooling water return line. Cooling water composition is as specified in the Process Design Basis.

- NOTES:**
- Maximum allowable sound pressure level for pump package : 85 dB(A) at 3 ft.
 - Minimum Design Metal Temperature = 10 °F
 - To be confirmed by vendor.
 - Vendor to limit the shut off pressure for the pump to 320 psig.
 - Seal plan type for pump to be confirmed by Rotary specialist / Vendor.
 - API standard 610 (latest edition) applies unless otherwise indicated.
 - NPSH available shall be at least 3 ft above NPSH required. NPSHa is specified at pump centerline assuming pump centerline is at 2.953 ft from grade.
 - Service is hazardous and non toxic.
 - Pump type: Horizontal. Pump in continuous operation.
 - Motor to be designed for end of curve operation considering start-up case as specified in Note-20.
 - Normal flow for the pump is for HMB Case 1 - Low Conversion SOR(Rev5).
 - Minimum flow for the pump is 50% of Normal flow.
 - Selected pump shutoff pressure including API margin shall not exceed estimated pump shutoff pressure indicated in datasheet.
 - Indicated flow (Rated / Normal / Minimum) is process required flow and does not include pump warm-up flow. Pump Warm-up line is provided with Restriction Orifice.
 - Vendor to confirm standby pump warmup-flowrate and shall include in specified pump rated capacity to arrive at final rated capacity of pump. Rated head shall be guaranteed without any -ve margin at final rated capacity of pump. Restriction Orifice on Warm-up line shall be loose supply by pump vendor.
 - Pump vendor to also advice minimum circulation flowrate for the pump.
 - Vendor to check and confirm with seal vendor, if steam jacketing is required for seal housing.
 - Maximum operating density is 42.82 lb/ft³.
 - Pumping temperature of the fluid ranges from 511.50 to 535.57 °F.
 - Viscosity of the fluid ranges from 0.24 t P.
 - Respiratory Expousre Hazard Class for P818 is R4.
 - Vendor to ensure that pump and all ass auxiliaries are suitable for following start-up condition:

Case	Flowrate (GPM)	Suction Pressure (psig)	Discharge Pressure (psig)	Differential Pressure (psi)	Differential Head (ft)	Available NPSH (ft)	Shaft Power (hp)	Density (lb/ft ³)	Viscosity (cP)	Temperature (°F)	Operation
Start-up	583	-3.1	176.6	179.8	632.5	26.5	43.7	40.94	0.22	541	Two (Parallel)

21. Vendor to provide the pump curve for parallel pump operation.

